Research

I study the use of data in education, especially in STEM education. My research spans two interconnected lines of work: using new data sources and computational methods to describe and question educational systems, and weaving data science into teaching and learning, with a focus on science education.

Publications Google Scholar CV

Educational Data Science

In the first line of work, I study educational systems using computational approaches and novel data sources. This includes research on educators' professional learning through social media, critical investigations of how K-12 institutions' social media practices compromise student privacy, and methodological work that integrates machine learning with qualitative analysis.

Data Science Education

In the second line of work, I develop and study approaches for data science learning in K-12 settings. This includes designing instruction for statistical reasoning — especially Bayesian thinking — in science classrooms, building tools and curricula that make data science accessible to newcomers, and contributing to the development of educational data science as a field through open-source software, a co-authored book (*Data Science in Education Using R*, Routledge, 2020), and a synthesis of what is known about K-12 data science learning.

Across both lines, I program in R and Python, use advanced statistical methods (especially multilevel models), develop open-source tools, and center partnerships with teachers. My current and near-future work extends these efforts into multimodal learning analytics, community-based data science learning, and the roles of generative AI in education.

Selected Projects

Project CREDIBLE

Creatively Reimagining Engagements with Data in Biology Learning Environments — an NSF CAREER-funded project on helping students learn with data in science classrooms.

Data Exploration and Sensemaking

An NSF-funded collaborative project on learning how to help middle grades science teachers integrate data exploration and sensemaking in the classroom.

Tools & Resources

- *Data Science in Education Using R* (Routledge) — a co-authored book with an open-access, web-based version accessed more than 150,000 times.
- Educational Data Science Graduate Certificate at the University of Tennessee, Knoxville.
- Open-source software and data resources, listed in my CV.

Funding

My research has been supported by more than $9 million in funding as principal investigator or co-principal investigator, primarily from the National Science Foundation, including an NSF CAREER award. A full list of grants is in my CV.